MIL-C-37366A
5 November 1976
SUPERSEDING
MIL-C-37366
2 January 1976

MILITARY SPECIFICATION

CATHETER AND NEEDLE UNIT, INTRAVENOUS, DISPOSABLE WITH SYRINGE (Needle Into Catheter)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sterile, nontoxic, pyrogen-free, disposable, radiopaque, catheter and needle intravenous placement units, with syringe. The units shall be suitable for administering intravenous medication from a single venipuncture in pre-operative to post-operative treatment.

1.2 Classification. Catheter and needle units shall be furnished in the following sizes as indicated below (see 6.1):

Size	- Catheter gage
A	- 14
B	- 16
C	- 18
D	- 20

Style	- Needle gage
a	- 16
b	- 17
c	- 18
d	- 19
e	- 20
f	- 21
g	- 22
h	- 23
i	- 24

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq., Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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2. APPLICABLE DOCUMENTS

2.1 Issue of documents The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein. These documents may be obtained as directed by the contracting officer.

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
MMM-A-250	- Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing).

MILITARY

MIL-S-36586	- Sterilization Strip Test, Bacterial Spore.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

No. 24380 CATHETER AND NEEDLE UNIT, INTRAVENOUS, WITH SYRINGE

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

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THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18024.)

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
Bureau of Biologics. Food and Drug Administration
Code of Federal Regulation Part 273, Minimum Requirements (as applicable).

(Application for copies should be addressed to the Food and Drug Administration, 5600 Fishers Lane, Rockville, MD 20852.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z70.1 *American Standard Dimensions of Glass and Metal Tapers for Medical Applications.*

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Plastic and rubber components. All plastic and rubber components coming in contact with intravenous medication and patient's blood shall pass the test for extractable metals when tested as specified in 4.4.1. The quantity of cadmium and lead shall not exceed 5 parts per million (p.p.m.) each, and the maximum barium content shall be 10 p.p.m. when tested as specified in 4.4.1.

3.1.1 Catheter. The catheter shall be fabricated from pliant fluorinated ethylene propylene (FEP Teflon) or polypropylene, of the best quality and suitable in all respects for the purpose intended. The catheter shall be radiopaque throughout its length. The finished catheter shall be sterile, pyrogen free, and shall meet the requirements of the USP for Class VI plastics, when tested as specified in 4.4.2.

3.1.2 Needle. The needle cannula shall be fabricated from drawn, or welded and drawn tubing; tubing shall be in accordance with GG-I-526, for class 3, type 304 material, and shall be subjected to the boil test as specified in GG-I-526.

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3.1.3 Catheter hub and needle hub. The catheter hub (when catheter is not of one-piece construction) and needle hub shall be fabricated from rigid or semi-rigid plastic, suitable in all respects for the purpose intended. The catheter and needle hub shall meet the requirements of the USP for Class IV plastics when tested as specified in 4.4.3. When the catheter is fabricated from one piece construction, the catheter hub shall be fabricated from rigid polypropylene and shall meet the requirements of the USP for Class VI plastics, when tested as specified in 4.4.2.

3.1.4 Syringe barrel. The syringe barrel shall be fabricated from rigid, transparent, polypropylene suitable in all respects for the purpose intended. The syringe barrel shall meet the requirements of the USP for Class IV plastics when tested as specified in 4.4.3.

3.1.5 Syringe plunger. The syringe plunger rod shall be fabricated from rigid polypropylene. The plunger rod shall meet the requirements of the USP for Class II plastics when tested as specified in 4.4.3.

3.1.6 Syringe plunger tip. The syringe plunger tip shall be fabricated from nonleaching natural rubber or neoprene of the highest quality, suitable in all respects for the purpose intended. The plunger tip shall meet the requirements of the USP for Class IV plastics when tested as specified in 4.4.3.

3.1.7 Needle guard. The needle guard shall be fabricated from a suitable plastic.

3.2 Style, design and dimensions. The catheter and needle intravenous placement unit shall be furnished complete with a syringe. The syringe shall be attached to the female hub cavity of the needle and shall be ready for use. The unit shall be suitable for administering intravenous medication from a single venipuncture in pre-operative to post-operative treatment. The attached syringe shall be suitable for obtaining sample of patient's blood while venipuncture is made. The unit shall consist of an outer radiopaque plastic catheter assembled over an inner hypodermic needle of short bevel design. The needle cannula shall make a safe, sliding fit within the outer plastic catheter. The catheter shall be permanently secured to a Luer-lock hub (Type I design), or as an alternate, the catheter shall be of one piece tapered construction (Type II design) with the proximal end of the catheter formed to a Luer-lock hub. The proximal end of the needle cannula shall be permanently secured to a needle hub. Needle hub shall be of the Luer-lock or Luer-slip type in accordance with Figure 3 or Figure 2, respectively, of ANSI Standard Z70.1. The outer edge at the distal tip of the plastic catheter shall be tapered to facilitate catheter insertion after venipuncture is made. The catheter hub shall form a Luer-lock connector in accordance with Figure 3 of ANSI Standard Z70.1, and in addition shall meet the requirements specified in 3.2.2. A needle guard shall be required. The catheter and needle gages specified in table I shall be furnished for the appropriate National Stock Number (see 6.3).

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TABLE I. Catheter and needle gages.

Catheter Gage Size	Needle Gage
A	a or b
B	c or d
C	e or f
D	g, h, or i

Style, design and dimensions shall be in accordance with DPSC Drawing 24380 Type I or Type II design, for the appropriate National Stock Number (NSN) and item identification.

3.2.1 Syringe. The syringe shall be of the Luer-lock or Luer-slip type. Syringe liquid capacity shall be 2-1/2 cubic centimeters (cc) or 3 cc. The syringe tip taper shall be in accordance with Figure 2, male end, of ANSI Standard Z70.1 and shall meet the requirements specified in 3.2.3. The syringe tip shall be formed concentric to the axis of the syringe barrel. Inside diameter of the barrel shall be concentric and designed to produce a liquid-tight system while simultaneously allowing free excursion of the plunger. The inside surfaces of the syringe barrel shall be uniform and free from pits, bubbles, mold marks, or other defects that would affect fluid-tightness and operation with the syringe plunger. The graduations of the syringe shall be 0.1 cc, with each 0.5 cc interval marked.

3.2.2 Diameter and depth of Luer hub cavities. The hub cavities of the catheters and needles shall be tested for diameter and depth of Luer hub cavity as specified in 4.4.4. Hubs which do not fit the plug gage within the "GO - NO GO" limits specified in 4.4.4 shall be rejected. In addition, when the hubs are fitted onto the plug gage, there shall be no movement of the hubs on the gage when the catheter and needle are pressed lightly from side to side in all directions.

3.2.3 Taper of syringe tip. The male Luer tip of the syringe shall be tested for proper taper as specified in 4.4.5. The external male Luer tapers which do not fit the ring gage within the "GO - NO GO" limits specified in 4.4.5, shall be rejected. In addition, when the ring gage is fitted onto the syringe tip, there shall be no movement of the gage on the syringe tip when the gage is pressed lightly from side to side in all directions.

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3.3 Security of assembly.

3.3.1 Catheter. Catheters of Type I design shall be tested for security of assembly. When tested as specified in 4.4.8, the catheter shall not loosen or pull out from the catheter hub. Catheters that yield or break during the test, but do not loosen or pull out from the hub, shall be considered to pass the security of assembly test.

3.3.2 Needle. When tested as specified in 4.4.8.1, the needle shall not break, loosen, or pull out of the needle hub.

3.4 Flexibility of catheter. When tested as specified in 4.4.9, the plastic catheter shall not crack or collapse. In addition, liquid flow shall not be obstructed in the internal passage of the catheter when tested as specified in 4.4.9.

3.5 X-ray opacity. When tested as specified in 4.4.13, the plastic catheter shall appear prominent and conspicuous when covered by at least 5/8 inch of aluminum.

3.6 Resistance to breakage of the needle cannula. When tested as specified in 4.4.10, the needle cannula shall not break.

3.7 Leakage of catheter. When tested as specified in 4.4.11, the connection of the plastic catheter with the catheter hub (Type I design units) shall not show leakage, and the Luer connection of the catheter hub (all designs) with the test gages shall not show leakage exceeding the formation of one clinging drop. As an alternate test, the above connections shall not show leakage exceeding the formation of one clinging air bubble when tested as specified in 4.4.11.1.

3.8 Leakage of syringe and needle. When tested as specified in 4.4.12, there shall be no leakage exceeding one falling drop between the syringe tip and needle hub or between the plunger tip and the syringe barrel.

3.9 Freedom from pyrogens. All components of the unit shall be free from pyrogens when tested as specified in 4.4.7.

3.10 Sterility. All components of the unit shall be sterile when tested with a biological indicator as specified in 4.4.6.

3.10.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National Stock Number.
c. Contract or purchase order number.

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d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i,e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel ,ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.10.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.2. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.11 Animal implantation test. The plastic catheter member only shall be tested, and shall show no evidence of toxicity or adverse tissue reaction upon gross and microscopic examination, when tested as specified in 4.4.2.1.

3.12 Instruction. A detailed instruction sheet for proper use of the catheter and needle unit shall be provided in each immediate container, or as an alternate with each package. A caution note on, or in, each immediate container shall state words to the effect:

"IF VENIPUNCTURE IS NOT SUCCESSFUL, WITHDRAW CATHETER COMPLETELY FROM PUNCTURE SITE BEFORE REINSERTING NEEDLE".

As an alternate, the caution note shall appear on the instruction sheet provided with each package.

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3.13 Certificates of freedom from pyrogens, biological tests, and animal implantation test. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested in accordance with methods described in the USP and found to be free from pyrogens, and have passed the biological tests and the animal implantation test. The certificate shall include the specific test performed and the test results. The record of the animal implantation test and its results shall be certified by the pathologist who performed or supervised performance of the animal implantation test. These certificates shall be submitted by the contractor in the manner as specified in 3.10.1.

3.14 Manufacturer's record. The manufacturer shall place a code or lot number on the immediate container which can be verified against the manufacturer's records of all tests made on sterile, pyrogen-free units. These records shall be made available for inspection by any duly qualified official of the Federal Food and Drug Administration. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.15 Compatibility. The materials of which the catheter and needle units are manufactured shall not react with or be acted upon, by the liquids with which they will be in contact during normal use and operation nor shall there be any material incorporated in the units which will cause any deleterious effect when the units are used for the purpose intended.

3.16 Workmanship. Workmanship shall be first class throughout. The end items shall be free from defects which detract from their appearance or impair their serviceability. The plastic catheter, needle cannula and syringe shall show a smooth finished outer surface free from pits, tool marks, burrs, scale, corrosion, and molding flash. Inner surfaces of all components of the unit shall be free from scale, deep pits, deep furrows and die marks. The lumen of the catheter, needle cannula, hub cavities and syringe shall be free from burrs, chips, dirt, slivers, imbedded particles, or other foreign matter. Needles shall be sharp, not blunt or bent, and points shall be free from feathered edges, burrs or other defects, when examined under 7 power (7X) magnification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the fluorinated ethylene propylene (Teflon), polypropylene, rubber and corrosion-resistant steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demostrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for examination purposes shall be one complete catheter and needle unit with syringe.

TABLE II. Sampling criteria for examination.

Type defect	Inspection level	AQL (percent defective)
Critical	III	1.0
Major	II	1.0

4.2.2 For Tests.

4.2.2.1 For sterility test. Sampling for sterility test shall be in accordance with the USP for parenteral devices and as specified in 4.4.6.

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4.2.2.2 For pyrogen test. Sampling for pyrogen test shall be as indicated in the USP for transfusion and infusion assemblies. Samples shall be selected which are to representative of each lot of plastic used in the fabrication of the units. As an alternate, sampling shall be on a sterilizer lot basis when sterilizer lots are smaller than, or equal to, production lots.

* 4.2.2.3 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105, and as indicated in table III. The unit of product for testing purposes shall be one complete catheter and needle unit with syringe. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE III. Sampling criteria for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
Needle Cannula	Composition analysis	3.1.2 & GG-I-526	GG-I-526	1/	--
	Boil test	3.1.2 & GG-I-526	GG-I-526	S-1	1.0
	Dimensions	3.2	--	S-1	2.5
	Security of assembly	3.3	4.4.8.2	S-1	0.65
	Resistance to breakage	3.6	4.4.10	S-1	1.0
Plastic Catheter	Extractable metals	3.1	4.4.1	2/	--
	Material - Class VI	3.1.1	4.4.2	2/	--
	Dimensions	3.2	---	S-1	2.5
	Security of assembly	3.3	4.4.8.1	S-1	0.65
	Flexibility test	3.4	4.4.9	S-1	1.0
	X-ray opacity	3.5	4.4.13	S-1	1.0
	Leakage test	3.7	4.4.11 or 4.4.11.1	S-1	0.65
	Animal implantation test	3.11	4.4.2.1	--	---
Catheter hubs and needle hubs	Extractable metals	3.1	4.4.1	2/	--
	Material - Class IV	3.1.3	4.4.3	2/	--
	Diameter and depth of Luer hub cavities	3.2.2	4.4.4	S-1	0.65

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TABLE III. Sampling criteria for tests.- Continued

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
	Extractable metals (all components)	3.1	4.4.1	2/	--
Syringe	Material- Class IV (Barrel and plunger tip)	3.1.4 and 3.1.6	4.4.3	2/	--
	Material - Class Class II (plunger)	3.1.5	4.4.3	2/	---
	Dimensions	3.2	---	S-1	2.5
	Taper of syringe tip	3.2.3	4.4.5	S-1	0.65
Syringe and needle	Leakage test	3.8	4.4.12	S-1	0.65

1/ Sufficient quantity of each lot of metal used in the construction of the needles shall be selected for chemical analysis.
2/ Sufficient samples to perform the extractable metal tests, and biological tests shall be selected from each lot and shall be representative of each batch of plastic and rubber (plunger tip) used in the manufacture of the catheter and needle units.

4.3 Examination.

4.3.1 Examination of catheter and needle units. The complete catheter and needle units shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables II and IV. Examination, however, shall not be restricted to defects listed in table IV. Any units in the sample containing one or more defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

TABLE IV. Classification of Defects.

Categories	Defects
Critical	
01	Catheter hub and needle hub not permanently secured to catheter and needle cannula, respectively.
02	Catheter surface pierced by needle.
03	Catheter bevel tip split, frayed, torn, incompletely formed, unevenly cut and not smoothly tapered.
04	Complete unit not free from foreign matter.

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TABLE IV. Classification of defects.- Continued

Categories	Defects
Major	
101	Catheter needle units not of the specified catheter gage and needle gage for the appropriate stock number.
102	One or more specified components missing.
103	Catheter surface not free from kinks, dents, pits, blisters, cuts, holes, and scratches.
104	Needle point not keen, not straight, and not free from feathered edges.
105	Needle surface not free from dents, pits, burrs, slivers, scale and corrosion.
106	Needle does not form a free, safe, sliding fit within catheter.
107	Instructions for use and caution note missing.
108	Syringe capacity less than 2-1/2 cc.
109	Heel of needle bevel does not protrude beyond distal end of catheter.
110	Syringe plunger tip not contained on plunger rod.
111	Syringe and syringe plunger not free from cracks, holes, breaks and flash.
112	Syringe orifice not complete or blocked.
113	Inside surfaces of syringe barrel not uniform, and not free from pits, bubbles and mold marks.
114	Syringe plunger not assembled into syringe barrel.
115	Syringe not assembled to needle hub.
116	Needle separated from catheter.
117	Needle guard does not cover catheter member and needle cannula member.

4.3.2 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Extractable metals. All plastic components (excluding needle guard) and rubber tip of syringe plunger shall be tested for extractable metals. The specimens shall be cut such that no thickness exceeds 3 millimeters (mm) and eluted for either 72 hours at 50 degrees Centigrade (^{o}C), 24 hours at 70° C or

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60 minutes at 121° C in two menstrua: 0.9 percent saline and cotton seed oil. Each gram of each specimen shall be covered with 5 milliliters (ml) of the menstruum (recommended amount is 10 grams of the specimen plus 50 ml of solvent). The menstrua shall be collected after agitation, at the end of the selected elution period and submitted to spectographic analysis for barium, cadmium and lead. The metals shall be reported in parts per million (p.p.m.) in the eluate. A value of 1.0 p.p.m. in the eluate shall be equivalent to extracting 1.0 milligram from 200 grams of test material in the elution period into 1000 ml elution menstruum.

4.4.2 Biological tests for Class VI plastics. The catheter shall be tested in accordance with the requirements of the USP for Class VI plastics.

4.4.2.1 Animal implantation test. Samples of the finished outer plastic catheter (end item) shall be tested in accordance with the USP for Implantation Test, Plastic Containers, for Class VI plastics, except after "72 hours" add "and not to exceed 144 hours"; and delete 'Examine macroscopically the area of the tissue surrounding the center portion of each implant strip. Use a magnifying lens if necessary,' and substitute: "'The muscle tissue surrounding the implants shall be examined, both grossly and microscopically, by a pathologist certified by the American Board of Pathology or the American College of Veterinary Pathologists'."

4.4.3 Biological tests for Class II & Class IV plastics. All hubs, syringe barrel and syringe plunger tip shall be tested in accordance with the requirements of the USP for Class IV plastics. The syringe plunger rod shall be tested in accordance with the requirements of the USP for Class II plastics; as an alternate, cottonseed extract may be used in place of 5 percent solution of alcohol in Sodium Chloride Injection specified in the USP.

4.4.4 Diameter and depth of Luer hub cavities. The Luer hub cavities of the catheter and needle shall be tested with the male Luer taper plug gage in accordance with Figure 5 of ANSI Standard Z70.1 (this gage shall be as shown in the ANSI Standard, except that it shall be constructed with a suitable liquid or pneumatic path to permit performance of the leakage test specified in 4.4.11). The female Luer cavities shall be slipped onto this gage with a straight motion, without turning or twisting the plug gage into the cavity. Once the female Luer cavity is slipped onto the gage, no additional force shall be exerted to force the plug gage into the Luer cavity. The fit of the Luer cavities onto this gage shall be tested by means of the feeler gage in accordance with Figures 5 and 6 of ANSI Standard Z70.1.

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4.4.5 Taper of syringe tip. The external tip of the syringe shall be tested with the female Luer ring gage in accordance with Figure 7 of ANSI Standard Z70.1. The syringe tip shall be slipped into this gage with a straight motion, without turning or twisting the syringe tip into the gage. Once the syringe tip is slipped into the gage, no additional force shall be exerted to force the syringe tip into the gage.

4.4.6 Sterility test. All components of the catheter and needle units shall be tested for sterility in accordance with the method specified in the USP for parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.4.6.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units.

4.4.6.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C
Dry Heat	B. stearothermophilus, or	55° - 65°C
	B. subtilis var.niger (or globigii)	30° - 35°C
Gas	B. subtilis var.niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.6.2 Inoculated carrier. Inoculated carriers (paper spore strips), if used, shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.6 Sterility test and 4.4.6.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.6.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

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4.4.6.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein, shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method.

4.4.6.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

4.4.7 Freedom from pyrogens. Samples of the finished units shall be tested for freedom from pyrogens in accordance with the USP for transfusion and infusion assemblies.

4.4.8 Security of assembly.

* 4.4.8.1 Catheter. The plastic catheter assembly of Type I design shall be tested for security of assembly by a device that utilizes the incline plane principle. The device shall apply a definite amount of load per unit of time (constant specimen-rate-of load). The device shall have suitable clamping mechanisms for securely holding the catheter hub and the plastic catheter proper. The device shall eliminate conditions such as specimen length, speed of operation, and specimen elongation qualities. The clamping device shall be placed on the plastic catheter proper as close to the hub-to-catheter junction as possible. This test shall be designed to determine the strength of the catheter hub to catheter junction and not the elongation characteristics of the catheter. The load specified in table VI, for the appropriate gage size, shall be the minimum load prior to any separation, or loosening, of the catheter to hub junction.

TABLE VI. Minimum load at separation or loosening.

Catheter size	Minimum load (pounds)
A	3.5
B	3.0
C	3.0
D	2.0

4.4.8.2 Needle. The security of assembly of the needle cannula shall be tested by applying a load of 10 pounds to the cannula and cannula hub for a period of at least 5 minutes, for needle gages styles a,b,c and d, with the cannula hub appropriately secured; and 5 pounds for at least 5 minutes for needle gages styles e,f,g, h and i.

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4.4.9 Catheter flexibility test. The plastic catheter shall be wrapped 90 degrees around a circular rod or mandrel, 1/2 inch in diameter. The condition of the plastic catheter shall be noted when released from the mandrel. A suitable liquid shall then be allowed to flow freely through the lumen of the catheter within 3 seconds under an internal pressure of 3 ± 1/2 pounds per square inch gage (psig).

4.4.10 Resistance to breakage. The hub of the needle cannula shall be clamped in a fixed position. The needle cannula shall then be bent from its original neutral position in a clockwise direction through an arc as specified in table VII for the appropriate needle gage, by applying a force to the cannula at a point (contact distance) as specified in table VII. A force shall then be applied in a counterclockwise direction (without changing contact distance) to bend the cannula past its neutral position to an angle specified in table VII. The cannula shall then be returned to its original neutral position. The test described herein shall be repeated 20 times.

TABLE VII. Resistance to breakage test criteria

Needle cannula gage style	Contact distance from hub (inches)	Bend (degrees)
a or b	1-1/4	20
c	1-1/4	25
d	1-1/8	25
e	1	25
f	3/4	25
g	11/16	25
h	5/8	25
i	1/2	25

4.4.11 Leakage test-catheter (hydrostatic method). With the catheter in a horizontal position, the catheter hub shall be inserted onto the plug gage specified in 4.4.4 (modified as specified in 4.4.4) to a water pressure source having a pressure gage. The catheter hub shall be inserted onto the plug gage with a straight motion, without turning or twisting the plug gage into the hub cavity with a force not be exceed 5 pounds. The feeler gage, specified in 4.4.4, shall be placed onto the plug gage to test that the "GO" position is not violated. Water shall be admitted to flow through the plastic catheter so as to fill the cavity space and the lumen of the catheter. The catheter shall then be sealed at its distal end in a leakproof manner, and a pressure of 20 psig shall be applied for at least 2 minutes.

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4.4.11.1 Leakage test - catheter (pneumatic method). This test shall be performed similarly to the test specified in 4.4.11, except that air at 10 psig shall be the testing media. The complete assembly of catheter hub to plug gage shall then be completely immersed into a suitable container of water, with the air pressure applied for at least one minute. The catheter assembly shall be observed for leakage as evidenced by air bubbles.

4.4.12 Leakage test of syringe and needle. The syringe tip shall be fully seated to needle hub. Water shall be drawn into the syringe, through the needle, to the top graduation mark (3 cc or 2-1/2 cc). The needle cannula point shall then be suitably plugged, an axial force equivalent of developing an internal pressure of 50 psig, shall be applied to the plunger rod for at least 10 seconds. The junctions of the syringe tip to needle and plunger tip to barrel shall be observed for leakage as specified in 3.8. This test shall be repeated with the plunger rod positioned midway between full and 50 percent capacity and at 50 percent capacity.

4.4.13 X-Ray Opacity test. The plastic catheter member only shall be placed on an unexposed X-ray plate and covered with an aluminum ladder ranging in thickness from at least 1/8 inch to 3/4 inch, in steps of 1/8 inch. X-ray exposure shall be as listed below; accomplished with automatic processing and using appropriate speed screens in a convenient size cassette:

Milliampere-second	(MAS) ----------	10
Peak kilovolts	(KVP) ----------	65 to 70
Focal length distance	(inches) ----------	30 to 40

The above stated exposure shall clearly delineate the steps 1/4 inch, 3/8 inch, 1/2 inch and 5/8 inch from each other.

5. PACKAGING.

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package containing fifty catheter and needle units, as specified, constitutes one unit of issue.

5.1.2 Level A.

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5.1.2.1 Individual container. Each catheter and needle unit shall be packaged in a sealed, peel open type, container or a rigid, plastic, tamper-proof container capable of maintaining sterility of contents unless damaged or opened. The peel open type container shall be designed with tabs to facilitate opening.

5.1.2.2 Unit package. Fifty individually packaged catheter and needle units of one size and style only along with instructions for use shall be packaged in a box conforming to PPP-B-636, class weather-resistant. The box shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Ten unit packages containing catheter and needle units of one size and style only shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the box specification.

5.2.2 Level B. Ten unit packages containing catheter and needle units of one size and style only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual container. In addition to any commercial markings, each container shall be marked with the sterility lot number, date of manufacture, name or trademark of contractor and the legend: "STERILE".

5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129 except that the date of manufacture shall be shown in lieu of the date packed. Sterility lot (control) number, contract or purchase order number and name of contractor shall be shown. The method of sterilization (G-Gas, R-Radiation, C-Cold, S-Steam, D-Dry Heat Sterilization) shall be shown following the level of packaging and the date of manufacture Example: "MFD-A 1/76-C" Final sterilization shall be on or after indicated date of manufacture.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129 except that the date of manufacture shall be shown in lieu

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of the date packed. The sterility lot (control) number and the date and method of sterilization as specified in 5.3.2 shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Applicable National Stock Number.

6.2 This specification is intended to cover the types which are normally procured to meet Military requirements. It does not cover all types, sizes, grades, etc., of the commodity indicated by the title nor all types which are commercially available.

6.3 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Specification Designation	
		Size	Style
6515-01-008-5209	CATHETER AND NEEDLE UNIT, INTRAVENOUS, With 2-1/2 or 3 ml Syringe, 14 Gage Catheter, Radiopaque, Disposable, 50s.	A	a or b
6515-01-008-5210	CATHETER AND NEEDLE UNIT, INTRAVENOUS, With 2-1/2 or 3 ml Syringe, 16 Gage Catheter, Radiopaque, Disposable, 50s.	B	c or d
6515-01-008-5211	CATHETER AND NEEDLE UNIT, INTRAVENOUS, With 2-1/2 or 3 ml Syringe, 18 Gage Catheter, Radiopaque, Disposable, 50s.	C	e or f
6515-01-008-7972	CATHETER AND NEEDLE UNIT, INTRAVENOUS, With 2-1/2 or 3 ml Syringe, 20 Gage Catheter, Radiopaque, Disposable, 50s.	D	g, h, or i

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* 6.4 Changes. The margins of this specification are marked with an asterisk to indicate where changes (additons, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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